

Neil Gandhi

412-506-0571 | npg5444@psu.edu | [linkedin.com/in/neilgandhi](https://www.linkedin.com/in/neilgandhi) | neilgandhi.github.io

EDUCATION

The Pennsylvania State University

Bachelor of Science in Mechanical Engineering

State College, PA

Fall 2027

– GPA: 3.67

– **Relevant Coursework:** Mechanical Vibrations, Manufacturing Processes, Engineering Materials, Circuits & Instrumentation, MatLab, Computational Tools, Thermodynamics I, Dynamics, Strength of Materials, ODE/PDE, Linear Algebra

EXPERIENCE

Research Assistant | Penn State Behrend

May. 2025 – Jan. 2026

- System Dynamics Modeling: Developed a 3-dimensional non-linear ODE model extending Lotka-Volterra equations to simulate interactions between one predator (Wolf Spider) and two prey species (Beetles and Grasshoppers).
- Stability Analysis: Applied Jacobian linearization, Center Manifold Theorem, and trapping regions to determine system equilibrium points and define conditions for species coexistence.
- Simulation: Utilized MATLAB to generate 3D phase-plane trajectories and numerical simulations, validating that populations remain bounded within specific invariant surfaces.
- Publication (Forthcoming): Co-author of "One Predator, Two Prey Lotka-Volterra Based Model" (Galiffa, Gandhi, et al.), proving mathematical time-reversal symmetry between 1-predator/2-prey and 2-predator/1-prey systems.

PROJECTS

Self-Balancing Ball & Plate System | Control Systems & RL

April. 2026 – Present

- Engineered a 2-DOF active stabilization platform, establishing a baseline closed-loop PID control architecture on an Arduino to maintain dynamic equilibrium of a steel ball.
- Developing a Reinforcement Learning (RL) framework using PyTorch and Stable Baselines3 (PPO) to train a continuous-action agent within a custom Gymnasium physics simulation.
- Iteratively deploying trained RL policies to the physical microcontroller to conduct comparative performance analyses between classical PID and machine learning control strategies.

Autonomous Line-Following Vehicle | Embedded Systems Project

Nov. 2024 - Jan. 2025

- Designed and constructed an autonomous vehicle utilizing an Arduino microcontroller and photoresistor arrays for closed-loop path tracking.
- Developed C++ control algorithms to process continuous sensor input and dynamically adjust DC motor PWM signals for real-time trajectory correction.
- Modeled and assembled a custom chassis to optimize sensor placement and ensure mechanical reliability during dynamic operation.

HackPSU Competition | Full-Stack Developer

Mar. 2026

- Collaborated in a 24-hour rapid development sprint to engineer and deploy a full-stack application.
- Developed and structured the backend architecture utilizing Python and the Flask framework.
- Identified and resolved real-time integration bugs between the backend and a Next.js frontend to ensure seamless user interaction.

LEADERSHIP & INVOLVEMENT

MATE UROV Competition | Lead Mechanical Contributor & Mentor

Aug. 2021 – June 2025

- Directed the mechanical design and systems integration for a hydrodynamic, neutrally buoyant Remotely Operated Vehicle, earning the Best Engineering Presentation award.
- Managed and led two multidisciplinary teams of 15+ students, coordinating the integration of mechanical, electrical, and propulsion subsystems for the RANGER and SCOUT competition classes.
- Collaborated cross-functionally to refine iterative designs, ensuring system reliability, manufacturability, and compliance with strict competition constraints.

Penn State Behrend Pickle Ball Club | Founder, President

Jan. 2025 – Dec. 2025

- Established a new student organization, successfully recruiting 30+ members across campus.
- Coordinate scheduling, facility management, and equipment for weekly practices, events, and tournaments involving 40+ participants.

HONORS & AWARDS

Honors: Behrend Honors Program (2024 – 2025)

Awards: The President Walker Award (2024)

TECHNICAL SKILLS & MEMBERSHIPS

Software & Tools: SolidWorks, Finite Element Analysis, Computational Fluid Dynamics, AutoCAD, Python, Fusion 360, MATLAB, LaTeX, C++

Professional Memberships: Nittany Motorsports (Formula SAE), Nittany Rocket Labs, Chess Club